Matías D. Molina | Curriculum Vitae

Personal Information

Name: Matías D. Molina.
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Passport: Argentine, Italian.
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Education

- **2015 - 2020/21** PhD in Computer Science. *Five-years degree*, Faculty of Mathematics, Astronomy, Physics and Computer Science, National University of Córdoba, Argentina. *In process of thesis-writing.*
- **2006 - 2012** Degree of Licenciado en Ciencias de la Computación. FAMAF-UNC.
  Overall Mark: 8 over 10. *Five-years degree. Equivalent to a BSc+MSc in CS.*
  Thesis: *Mathematical Morphology for Image Segmentation.* Dissertation Mark: 10 over 10.

Skills & tools

Machine learning | Statistics | Zero-shot learning | Teaching | Python | PyTorch | NumPy | SciPy | Scikit-learn | Pandas | Matplotlib | Seaborn | Gnuplot | Git | LaTeX | Friendly | Creative | Empathic | Communicative | Responsible

Projects and Publications

- Analyzing data granularity and semantic structures in Zero-Shot Classification
- Exploring Projection Symmetries in Zero-Shot Classification (sent to Pattern Recognition Letters)
- Performance Variability on Zero-Shot Classification (accepted at NeurIPS'20)
- Molina M., Sánchez J. (2018) Zero-Shot Learning with Partial Attributes. In: Brito-Loeza C., Espinosa-Romero A. (eds) Intelligent Computing Systems. ISICS 2018. Communications in Computer and Information Science, vol 820. Springer, Cham.

Scholarships

- **2015** PhD scholarship provided by the National Council of Technical and Scientific Research (CONICET).
- **2011** TICs Scholarship provided by The National Ministry of Education for undergraduate students.

Conferences and Workshops

- **2020** Presenter at LXAI workshop at NIPS.
- **2020** Attendence to BMVC, CVPR, NIPS.
- **2020** Attendence to MachinesCanSee 2020 http://machinescansee.com/.
- **2018** Presenter at the International Symposium on Intelligent Computing Systems 2018, Mérida, México. *Zero-Shot Learning whit Partial Attributes.*
- **2017** Attendance to the Workshop *Hands On Data*, Catholic University of Córdoba, Argentina.
- **2017** Attendance to the *XLIII CLEI – 46 JAIIO, Argeninean School of Artificial Intelligence*. UTN, UNC. Córdoba, Argentina.
- **2017** Attendance to the Workshop *Publishing your research*. Faculty Chemistry Science, UNC, Argentina.

University tasks

- **2017,2018** Member of the teacher's assistants evaluation court at the National University of Córdoba.
- **2018,2019** Computational thinking workshop. Organized jointly with Dra. Emilia Echeveste

Courses

Postdegree courses

- **2016** (i) Machine Learning and Computer Vision, (ii) Stochastic Classification (FAMAF – UNC).
- **2015** (i) Non-linear Optimization, (ii) Introduction on Computer Vision and GPUs (FAMAF – UNC).
- **2010, 2011** Artificial Intelligence, Statistical Analysis on Satellite Imagery, Computer Graphics (FAMAF – UNC).

Jobs and Teaching

Teaching

- **2014 - present**: Teaching Assistant FAMAF - UNC
  - **2020**: Introduction to Algorithms.
  - **2019, 2018, 2016**: Introduction to Algorithms. Operative Systems.
  - **2017**: Paradigms of Programming Languages, Operative Systems
  - **2015**: Networks and Distributed Systems, Operative Systems.
  - **2014**: Computer Organization, Operative Systems.
- **2012**: Teacher of Informatics, {1st. 3th.} Level. UPAMI. Secretary of Extension, UNC.*In this position I worked in an interdisciplinary team together with psychologists and social communicators, scheduling informatic courses to the general public.*
- **2009**: Undergraduate Teaching Assistant. FAMAF - UNC.

Others

- **2017** LAB design for the Paradigms of Programming subject.
- **2018,2017,2016,2015** LAB design for the Operative System subject.
- **2015** Co-advisor of Licentiate thesis under the supervision of Professor Dr. Oscar Bustos.

Software Development

- **2015 – 2020** Python + PyTorch. FAMAF, UNC.*During my PhD I've been reproducing, designing and implementing different machine learning models. Mainly based on Python, using different libraries for statistics: NumPy, SciPy, SiKit-learn, Matplotlib, Seaborn, Pandas, etc.*
- **2014 – 2015** Python Developer. FAMAF, UNC.*Document management application.*
- **2013 – 2014** Python Developer. Company: Harriague + Asociados. Córdoba, Argentina. *Design and development. SCRUM / REST / Quality Control/ Requirements Specification*

Languages

Spanish: Native - English: Intermediate - Portuguese: Intermediate.